PROBLEMS WITH PLASTICS? 3

Plastics have been described as being 'too good'. They have become popular because they:

- are made from relatively cheap raw materials (petrochemicals)
- can be moulded and shaped or drawn into fibres
- are tough
- can be coloured
- are good thermal and electrical insulators
- have good safety and hygiene properties for packaging food and medicines
- are lighter than alternatives
- are resistant to chemicals.

The world's annual consumption of plastic materials has increased from around 5 million tonnes in the 1950s to nearly 100 million tonnes today. Plastics degrade very slowly. No one knows how long they take to break down because plastics have only been in existence for a relatively short time. Degrading slowly is a useful property while plastics are being used, but a problem when plastics are discarded. The disposal of plastics causes problems; e.g. they can give off toxic fumes when burnt. Many people think that less plastic should be used. Other people claim that plastics are an essential part of modern living; e.g. the use of plastics has made cars lighter and more fuel-efficient. There are ways that the amount of plastic thrown away can be reduced.

- Use less plastic; e.g. use alternative (conventional) products
- Recycle
- Develop biodegradable plastics.

The standard supermarket carrier bag is made out of LDPE. One estimate is that over 200 bags per person are given out and discarded in Australia every year. Many retail businesses are trying to discourage customers from using plastic bags by charging for them and by suggesting alternative ways of carrying goods.

1 What does the acronym *LDPE* stand for?

2 Why would a retailer want to reduce the number of plastic bags used by their customers?

3 At one supermarket, they decide to put customers goods in large paper bags instead of the usual plastic bags. Outline the possible advantages and disadvantages of this decision.

4 At another supermarket, a bin is provided in which people can put their old plastic bags to be recycled later. Outline the possible advantages and disadvantages of this.

'Green' plastics There are efforts to increase the amount of plastic being recycled, but there are problems with recycling plastic. Thermoplastics can be melted and reused but thermoset plastics cannot. The many different types have to be separated before recycling can begin, and this is very labour intensive. In addition, the value of the material is low so recycling plastics is not very profitable. Some vehicle manufacturers are modifying their designs to make the plastic parts easier to recycle when the vehicle reaches the end of its useful life. It is estimated that making plastic carrier bags from recycled material uses 35% of the energy and 10% of the water required to make bags from newly-manufactured polyethylene. It also produces about 50% of the pollution.

5 Clothes often come in clear plastic bags marked: *Not a toy. Keep away from children under the age of three*. Why do the plastic bags carry this warning?

6 Outline the advantages and disadvantages of making plastic carrier bags from recycled rather than 'virgin' polyurethane.

Scientists have developed polymers that break down. These can be **biodegradable** or **photodegradable**. Biodegradable plastics will degrade in landfill but can release methane – a greenhouse gas. Photodegradable plastics only break down on exposure to light.

Biopolymers have been developed from renewable resources. Genetically-modified organisms such as plant sugars can be used to make plastics. These plastics are still expensive and in the early stages of development. Some of the more important biopolymers are starch-based. This is because starch is easy to obtain, affordable, renewable and inexpensive to process as a biopolymer.

7 Will photodegradable plastics break down in landfill? Explain your answer.

8 Outline the differences between a *biodegradable polymer* and a *biopolymer*.

9 Outline why plastics have become popular.

10 Name a naturally-occurring substance that can be used to make a biopolymer.

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